Project Backlog

ID	Epic / User Story	Effort	Priority	Sprint	Owner	Status
E0	**Project Planning, Requirements & Design**		**High**	S1/S4	Team	In Progress
T0.1	Define the project vision, initial scope, team roles, and initial product backlog.	2	High	S1	Team	Complete
T0.2	Set up GitHub, branching strategy, project board, and team development workflow.	3	High	S1	Team	In Progress
T0.3	Complete and refine the project proposal and semester project plan.	3	High	S2	Team	In Progress
T0.4	Define functional requirements and acceptance criteria for the core application.	5	High	S3	Toby	Not Started
T0.5	Complete a feasibility study for Swift/SwiftUI, data storage, authentication, AI, and pose estimation.	5	High	S3	Carlos	In Progress
T0.6	Create wireframes and user flows for login, home, skill tree, lessons, and progress screens.	3	High	S3	Turner	Not Started
T0.7	Design the initial software architecture, data model, and major app components.	5	High	S4	Team	Not Started
E1	**User Accounts & Core App Navigation**		**High**	S4/S5	Team	Not Started
U1.1	As a user, I want to create an account so that my progress can be saved.	3	High	S4		Not Started

U1.2	As a user, I want to log in and log out so that I can securely access my account.	3	High	S5		Not Started
U1.3	As a user, I want the app to remember my login so that I do not need to sign in every time.	3	Medium	S5		Not Started
U1.4	As a user, I want clear navigation between the home screen, skill tree, lessons, and profile so that the app is easy to use.	3	High	S4		Not Started
U1.5	As a new user, I want to choose the skill I want to learn so that I can begin the appropriate skill tree.	3	High	S5		Not Started
E2	**Skill Tree Framework**		**High**	S4/S5	Team	Not Started
U2.1	As a user, I want to view a visual skill tree so that I understand how lessons are organized.	5	High	S4		Not Started
U2.2	As a user, I want to select a lesson from the skill tree so that I can begin learning that topic.	3	High	S5		Not Started
U2.3	As a user, I want future lessons to remain locked until I complete their prerequisites so that I follow the intended progression.	5	High	S5		Not Started
U2.4	As a user, I want completed lessons to be marked so that I can easily see my progress.	3	High	S5		Not Started
U2.5	As a user, I want my lesson completion to remain saved after closing the app so that I do not lose my progress.	5	High	S5		Not Started
U2.6	As a user, I want to return to previously completed lessons so that I can review material.	2	Medium	S6		Not Started

E3	**Gym & Fitness Skill Tree**		**High**	S5/S6	Team	Not Started
U3.1	As a user, I want to learn the correct steps for performing the bench press so that I understand the exercise.	3	High	S5		Not Started
U3.2	As a user, I want visual examples and explanations of proper form so that I know what the movement should look like.	3	High	S5		Not Started
U3.3	As a user, I want to learn common bench press mistakes so that I know what errors to avoid.	2	Medium	S5		Not Started
U3.4	As a user, I want general starting weight and progression guidance so that I can progress through the lesson structure.	3	Medium	S5		Not Started
U3.5	As a user, I want to record the weight and repetitions I performed so that I can track my training history.	5	Medium	S6		Not Started
U3.6	As a user, I want a dashboard displaying completed lessons and progress so that I can see how I am improving.	5	High	S6		Not Started
E4	**Pose Estimation & AI Form Feedback**		**High**	S3/S7	Team	Not Started
T4.1	Research and prototype a pose estimation solution capable of detecting body joints from an image or video.	5	High	S3		Not Started
T4.2	Create a basic prototype that verifies pose estimation works with a bench press movement.	5	High	S4		Not Started
U4.3	As a user, I want to record or upload a short exercise video so that the app can analyze my form.	5	High	S5		Not Started

U4.4	As a user, I want the app to identify important body joints during my exercise so that my movement can be evaluated.	8	High	S6		Not Started
U4.5	As a user, I want the app to calculate important joint positions or angles so that it can identify possible form problems.	8	High	S6		Not Started
U4.6	As a user, I want feedback explaining possible form mistakes after my video is analyzed so that I know what to improve.	8	High	S6		Not Started
U4.7	As a user, I want AI-generated explanations of my form feedback so that the recommendations are easier to understand.	5	Medium	S7		Not Started
U4.8	As a user, I want form analysis to occur while I exercise so that I can receive feedback with minimal delay.	8	Low	S7		Not Started
E5	**Quality, Testing & Non-Functional Requirements**		**High**	S5/S7	Team	Not Started
T5.1	Define measurable non-functional requirements for usability, performance, reliability, and error handling.	3	High	S5		Not Started
T5.2	Create test cases for account creation, login, progress saving, skill tree unlocking, and lesson completion.	5	High	S6		Not Started
T5.3	Implement unit tests for core application logic.	5	High	S6		Not Started
T5.4	Test integration between the user interface, stored progress, gym lessons, and pose estimation system.	5	High	S7		Not Started

T5.5	Improve loading states, error messages, usability, and performance based on testing results.	3	Medium	S7		Not Started
T5.6	Fix identified bugs and refactor code that is difficult to maintain.	5	High	S7		Not Started
E6	**Final Integration & Semester Deliverable**		**High**	S8	Team	Not Started
T6.1	Perform final regression testing of the complete application.	5	High	S8		Not Started
T6.2	Confirm all required high priority user stories meet their acceptance criteria.	3	High	S8		Not Started
T6.3	Create a stable demonstration build of the iPhone application.	5	High	S8		Not Started
T6.4	Prepare the final demonstration, presentation, screenshots, and explanation of the development process.	3	High	S8		Not Started